

Visualizing System Logs with Splunk: From Setup to Custom Dashboards

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Overview

This article walks through the initial setup and configuration of **Splunk**, a powerful tool for log management and system monitoring. The focus is on installing Splunk, clearing Windows Event Viewer logs for a clean start, and building a basic dashboard to visualize new system events in real-time.

Project Goals

This project was designed to:

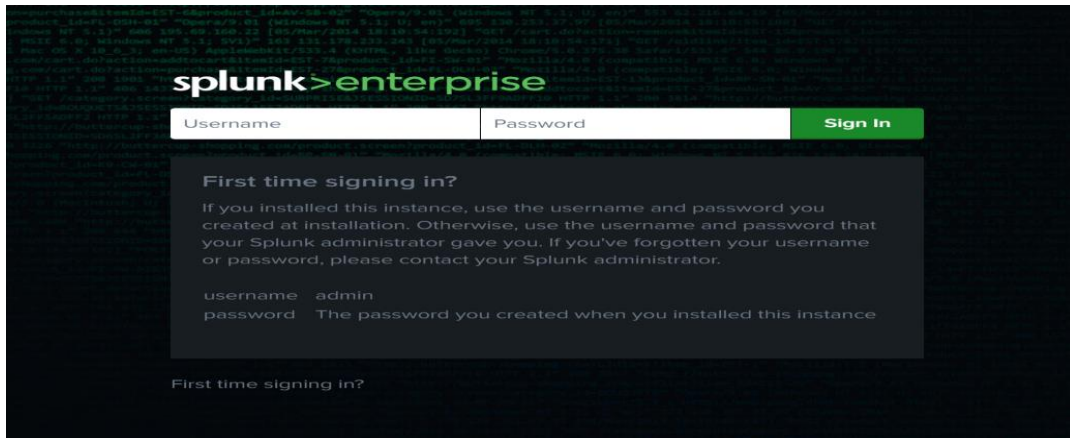
- Set up **Splunk** locally on a Windows machine
- Clear existing logs from **Windows Event Viewer**
- Create a **custom Splunk dashboard** to monitor system activities using targeted search queries

By doing this, the environment was prepared for more accurate log analysis and system behavior tracking moving forward.

Methodology

1. Installing Splunk

- Visited the [official Splunk website](#)
- Downloaded **Splunk Enterprise**, suitable for Windows
- Created a Splunk account to access the download
- Installed the software and set up admin credentials
- Launched Splunk via the Start menu and logged in successfully

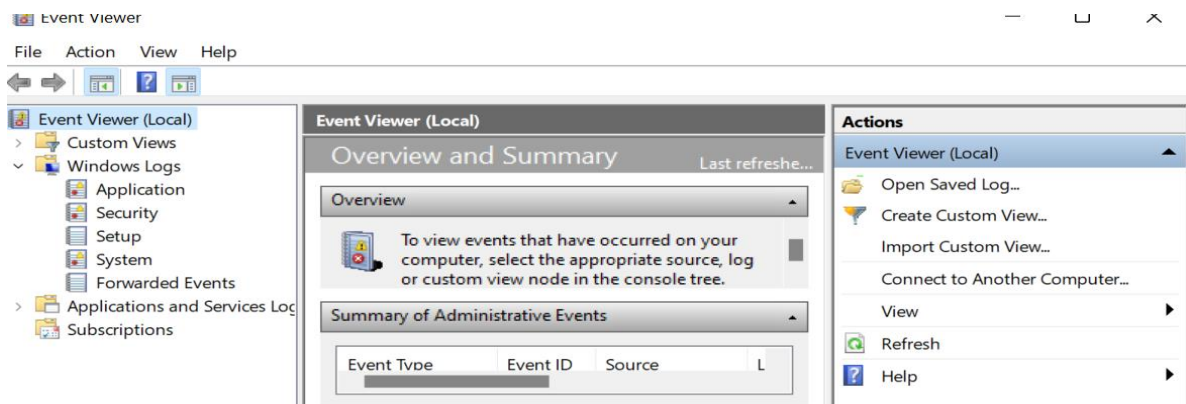


First login to Splunk using newly created admin credentials after successful installation on the local system.

2. Clearing Windows Event Logs

To ensure a fresh log monitoring environment:

- Opened **Event Viewer** via the Start menu
- Navigated to *Windows Logs* → selected categories like *Application*, *System*, or *Security*
- Chose “Clear Log” (with or without saving)
- Cleared logs to prepare for new data collection

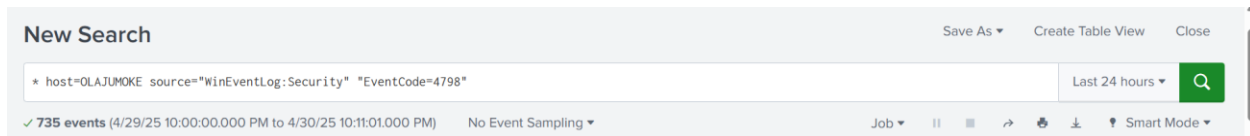


Windows Event Viewer used to clear existing system logs creating a clean environment for accurate monitoring in Splunk.

3. Using Splunk for Monitoring

- Accessed Splunk's **Search & Reporting** feature
- Ran custom search queries, such as:

*** host=OLAJUMOKE source="WinEventLog:Security" "EventCode=4798"**

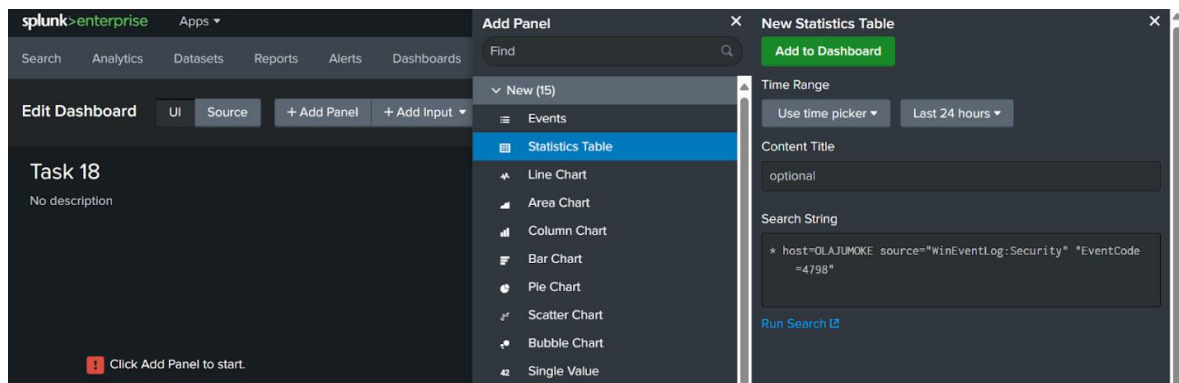


A Splunk search query targeting security events for live log filtering and analysis.

- Used the query results to build a dashboard panel

4. Creating the Dashboard

- Clicked on **Dashboards** → *Create New Dashboard*
- Named the dashboard and chose **Classic Studio**
- Added a new panel (e.g., table or graph)
- Pasted the search query and selected a time range (e.g., last 15 minutes)



Creating a new Splunk dashboard panel using Classic Studio for visualizing filtered log data in real time.

- Adjusted the layout and saved the dashboard

#	_time	host	source	sourcetype
>	3/27/25 2:39:30.000 PM	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security
>	3/27/25 2:39:29.000 PM	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security
>	3/27/25 2:38:57.000 PM	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security
>	3/27/25 2:38:57.000 PM	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security
>	3/27/25 2:37:13.000 PM	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security
>	3/27/25 2:37:13.000 PM	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security
>	3/27/25 2:34:44.000 PM	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security
>	3/27/25	OLAJUMOKE	WinEventLog:Security	WinEventLog:Security

Custom Splunk dashboard showing a visual summary of Windows Security log events.

Key Insights

Setup Success

- Splunk was installed without issues
- Logs were successfully cleared
- A working dashboard was created with visual feedback on live system activity

No Threat Indicators Found

Since the focus was on setup and visualization—not threat detection—no Indicators of Compromise (IOCs) were observed during this phase.

Impact and Benefits

- **Log Visibility:** Splunk provides centralized access to event logs across categories (system, application, security)
- **Pattern Detection:** Visualization helps highlight trends, anomalies, or errors in log data
- **Preparedness:** The system is now better equipped for future threat hunting or compliance monitoring

Recommendations

Immediate

- No critical actions required. The current setup is stable and functional.

Long-Term

- Expand the dashboard with more panels for different log sources
- Set up **scheduled searches** and **alerts** to detect suspicious behavior automatically

Conclusion

Splunk is now active and configured, with a clear dashboard displaying filtered logs from Windows Event Viewer. This foundation sets the stage for more advanced monitoring, alerting, and threat detection capabilities.

Splunk was successfully installed and configured, and a dashboard was created using search queries. Clearing the logs in Event Viewer helped provide a clean slate for accurate monitoring. With the basic setup complete, the system is now better prepared for ongoing event analysis.